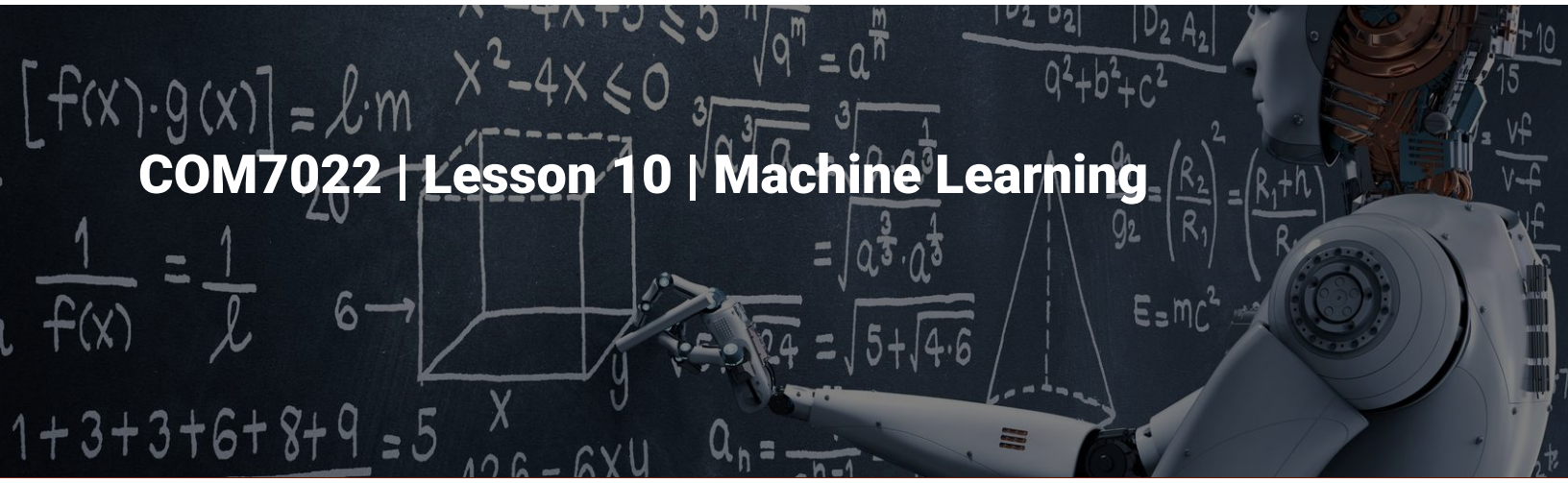


COM7022 | Lesson 10 | Machine Learning

Machine Learning (COM7022)

Lesson 10: Summary

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- Lesson introduction
- Before you get started
- Topic 1: Summary of Lessons
- Topic 2: Assignment support
- Lesson summary
- Reading List

Lesson introduction

Welcome to Lesson 10

LESSON INTRODUCTION | LEARNING OUTCOMES

Welcome to Lesson 10. The FINAL LESSON!

Congratulations on almost completing the Machine Learning module. This week we will be concerning ourselves with reflection of the module in its entirety. We will begin this reflection week by summarising each individual lesson and contents within. We will then discuss the many different departments that are at Arden to help support and guide you through not only your studies, but student life in general.

So, let's begin the lesson with the summarised content of each lesson we have completed.



Computing

At Arden University

LESSON INTRODUCTION

LEARNING OUTCOMES

By the end of Lesson 10, you will not have only completed the module in its entirety, but you will have also have:

- A summary of each individual lesson
- Detailed knowledge where you can access additional support throughout the module and your wider degree



Computing

At Arden University

Continue to Before you get started

Before you get started

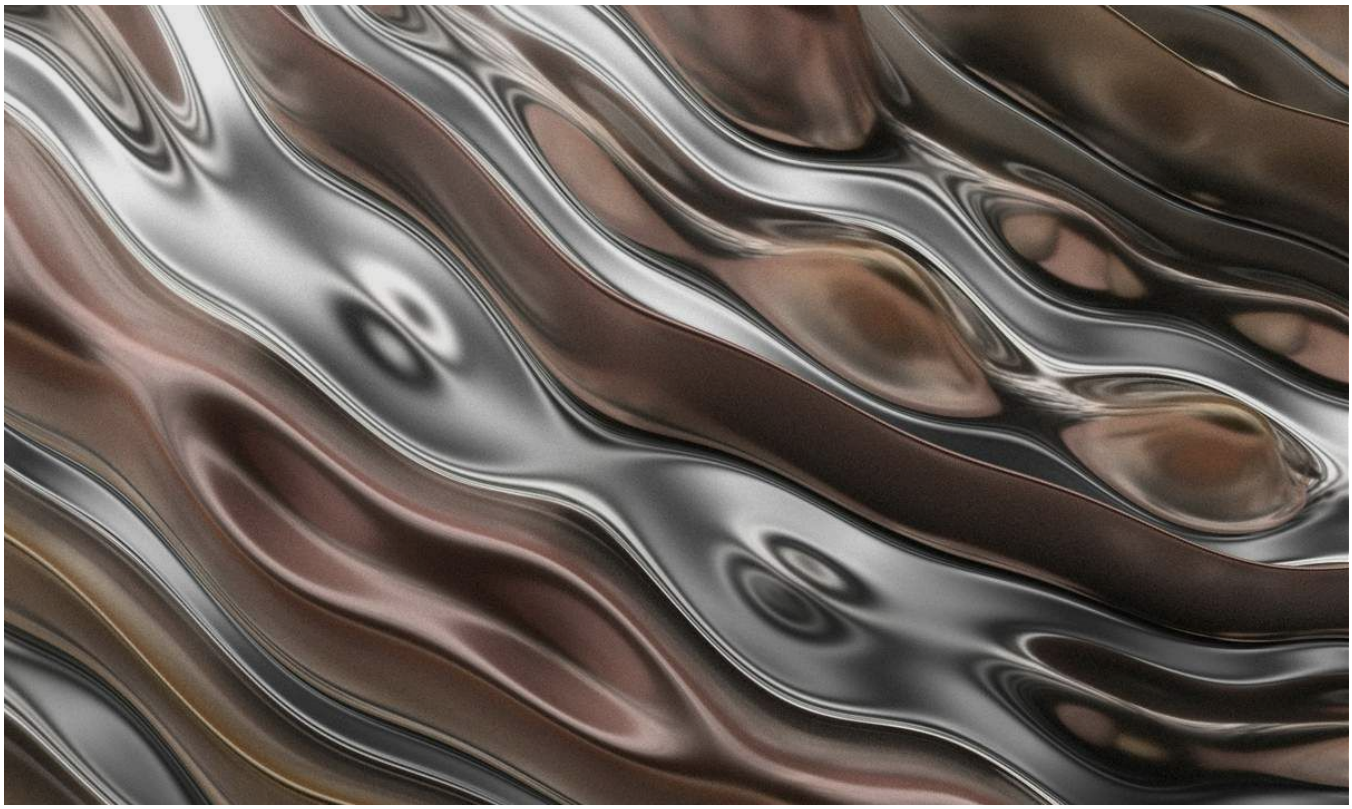


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Did you know?

Arden University was a finalist for seven EducationInvestor Awards in 2023 (Arden University, 2023a).

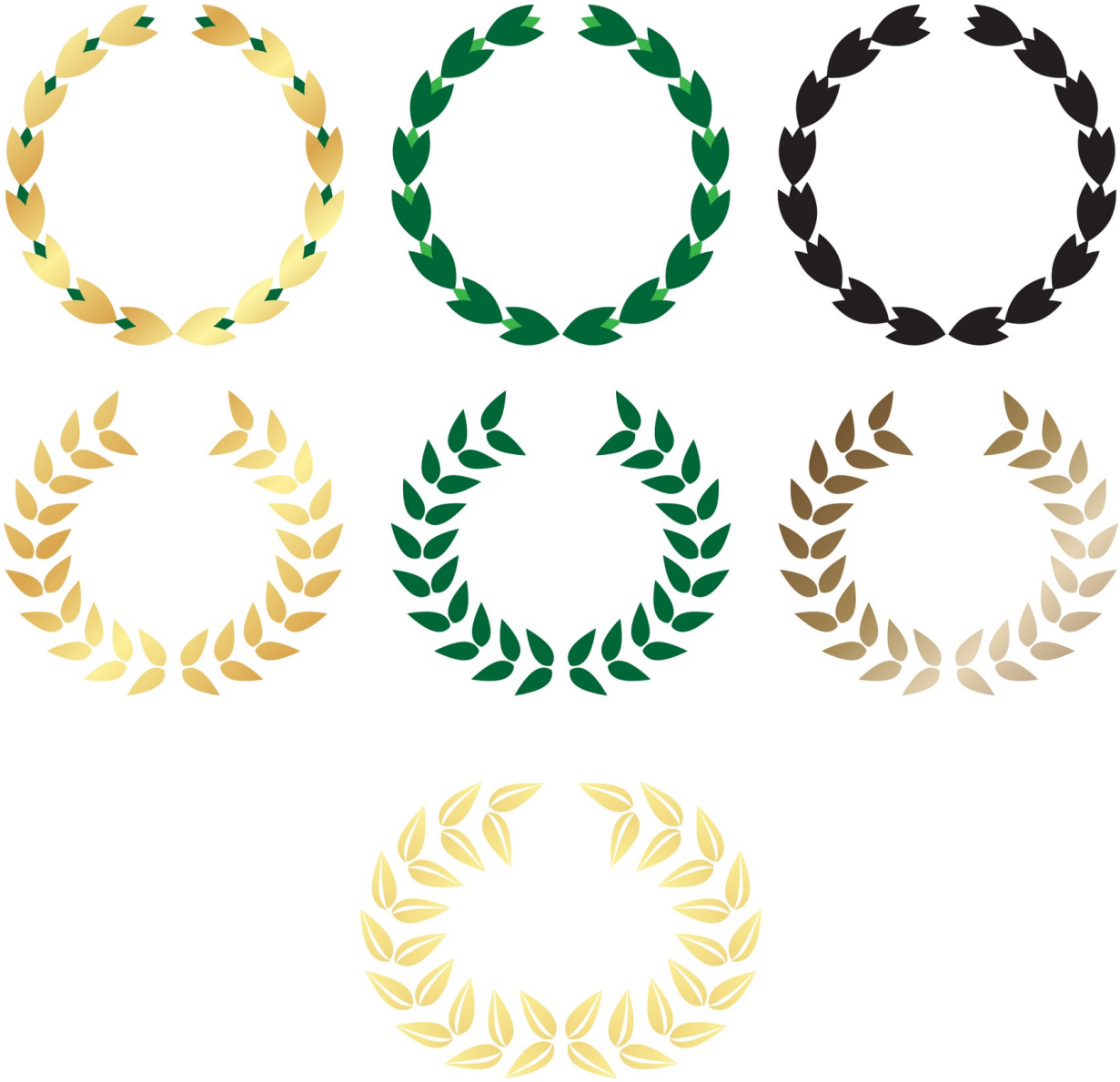


Image from Articulate Rise 360 Content Library.



Image from Articulate Rise 360 Content Library.

Arden University focuses on skills for employability within their degrees (Arden University, 2025).

Continue to Topic 1

Topic 1: Summary of Lessons

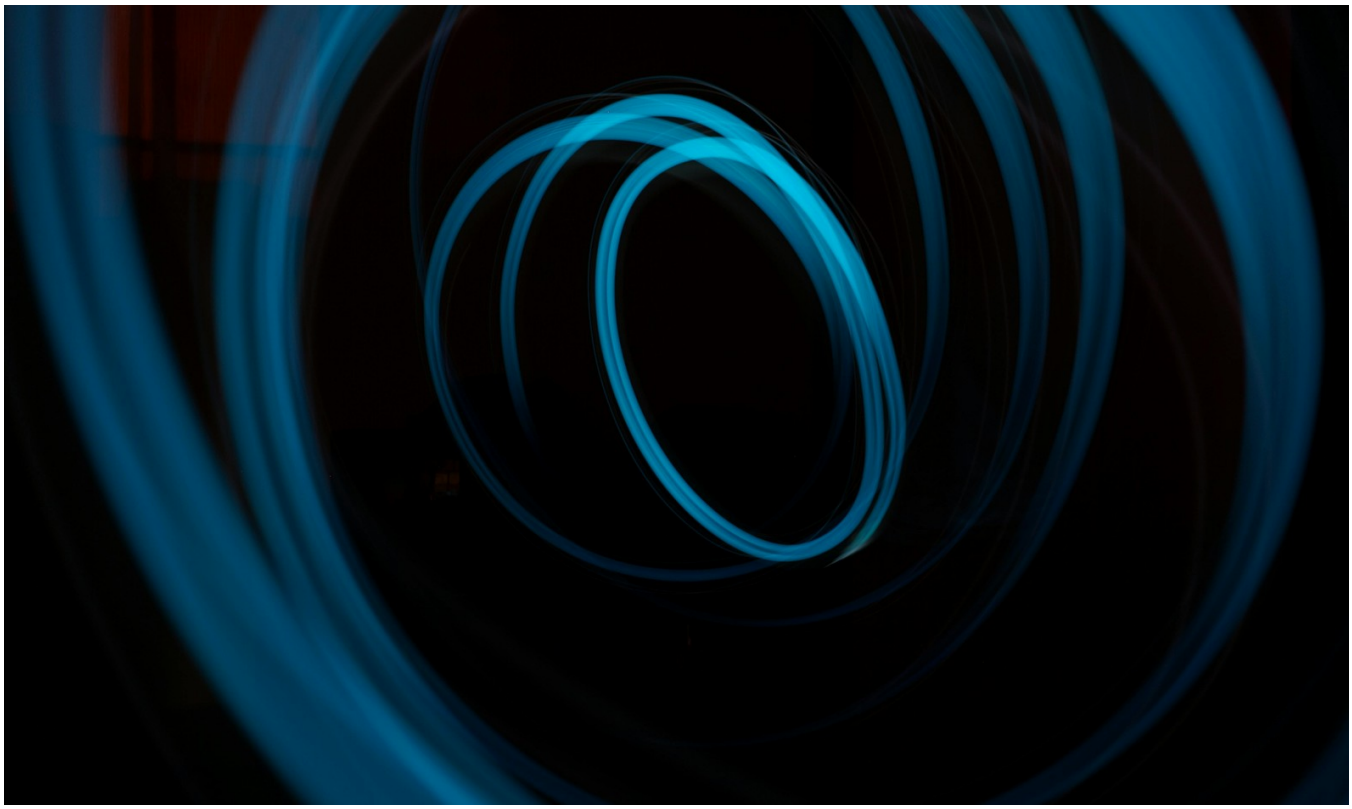


Image from Articulate Rise 360 Content Library.

Lesson 1

Lesson 1 was an introduction into the world of machine learning for data scientists, discussing the meaning of ML and the challenges it poses to an ML practitioner. In the first topic, we introduced the concept of ML and discussed the type of learning and real-world applications of ML. Topic 2 explored the essential toolbox that every budding ML analyst should have. Some of these individual tools you may already be confident with, and some not. Either way, recapping and solidifying fundamental knowledge are essential. Topic 3 provided a general review of two

packages, NumPy and pandas. These are the fundamental libraries in Python when dealing with data, which is an essential part of becoming an ML practitioner and data scientist. This topic also links what we have learned in our previous module, COM7024 – Programming for Data Analytics, and how we can contextually build our skills onto machine learning environments. Finally, in Topic 4 we discussed the main packages, Scikit-learn and SciPy, that we will use in this module for developing our hands-on practical skills and for building on the ML algorithms we will be learning.

Continue

Lesson 2

Raw data is a constant worry for a data scientist. Sometimes, when we receive the raw data, it is quite normal to be shocked by the state of the data. Hence, data scientists usually spend about 80% of their time on the processes we discussed in this lesson: data cleaning, data integration, data reduction and data transformation.

After the first stage of cleaning the data, removing or replacing duplicate, redundant and missing data, we looked at some techniques of data integration. We then focused on compressing and condensing our data meaningfully without the risk of losing the information and quality of the data. Finally, we discussed data transformation and how this process allows us to convert our data into a better representation and even to create new attributes that might enhance our data quality.

Continue

Lesson 3

In this lesson, we explored regression, from the intuition of linear regression to LASSO regression for complex, multiple-independent variables/features datasets.

Continue

Lesson 4

Lesson 4 began with some theoretical concepts that we needed to understand to carry out association rule mining. In addition to understanding the purpose of association rule mining, we also looked at the key metrics of association rules: the support, confidence and lift of the association rules. We saw the formulas for calculating these metrics which we must understand to appreciate the metrics and how to use them.

We then moved onto the Apriori algorithm, which is a commonly used algorithm for finding association rules, as it aids in making the search efficient by eliminating item sets that are not frequent enough to be worth considering.

Continue

Lesson 5

The first topic focused on decision trees. We also demonstrated information gain, both theoretically and numerically. We initially gave a detailed demonstration of how a decision tree algorithm works and the improved efficiency process involved in producing, as far as possible, the optimum model.

In Topic 2, we looked at an improved machine-learning technique in the information gain process demonstrated in the first topic, namely, gain ratio. Hopefully, Topic 2 has shown you

that trade-offs can be expected with a complex process, and gain ratio is no exception.

Topic 3 saw the humble classification tree turned into a forest of trees. Random forest algorithms are widely used within machine learning for obvious reasons. Hopefully, you can now determine why a collection of decision trees is much better than a singular classification decision tree.

In the final topic, we looked at pruning, which is a technique for improving the accuracy and efficiency of a decision tree algorithm.

As we extensively demonstrated, pruning is a machine-learning technique used to optimise the efficiency of a decision tree; in other words, it limits overfitting and underfitting within the training data. Moreover, we also discussed how pruning aids in reducing the size of the decision tree by removing elements that are not really relevant to the model. However, a balance must be sought between complexity and simplicity.

Continue

Lesson 6

In this particular lesson we looked at and demonstrated the Naïve Bayesian algorithm and ideology. The first topic addressed what Naïve Bayesian is and how in what machine learning environment it can be implemented. Having given a solid foundation in which to build we progressed forward to Topic 2 where we looked deeper into the methodology and how it can be used to attain realistic results. In Topic 3 we gave a full example on how to build, apply and gain results. Admittedly, the model was a touch challenging, but nothing we could not do again and apply to an alternative data set. Finally, in Topic 4 we looked at the Bayesian Belief Network.

Continue

Lesson 7

Without doubt, full SVM and SVR studies are involved. However, within Topic 1 we gave the ideology of SVM, including graphical representation. SVM is not only limited to 2-D and can be applied in multi-dimensions, although when higher dimensions are sort, the process can become quite complex, quite quickly. Topic 2 developed on from the theory given within Topic 1. Here a full SVM investigation was given, with comments within the code.

Within Topic 3 mainly focused upon the other Support Vector ideology, in other words SVR. The final topic summed up both SVM and SVR: very similar, just quite different. Hopefully, having concluded the lesson this has become clear.

Continue

Lesson 8

Lesson 8 was the lesson on K – Means Cluster analysis.

In Topic 1 we detailed theoretical explanations of what clustering is and more importantly why it is used so extensively within a machine learning environment. Although cluster analysis would not be used as a standalone technique, it is nevertheless a very good starting point on which to impose upon an unlabelled data set. Topic 2 discussed the most common clustering techniques and their differing associated approaches. We initial began by determining the clear difference between hard and soft clustering and in doing so, allowed us to progress further into Hierarchical clustering, density-based clustering and the overlooked clustering technique, Grid clustering.

Within Topic 3, we focused on arguably the most common type of clustering algorithms, K – Means. As we are now aware, K – Means is a portioning clustering technique and as such, pre-define centroids are input into the model, along with the distance from each centroid to the data points.

In the final topic, we demonstrated a K – Means cluster investigation on a targeted data set.

Continue

Lesson 9

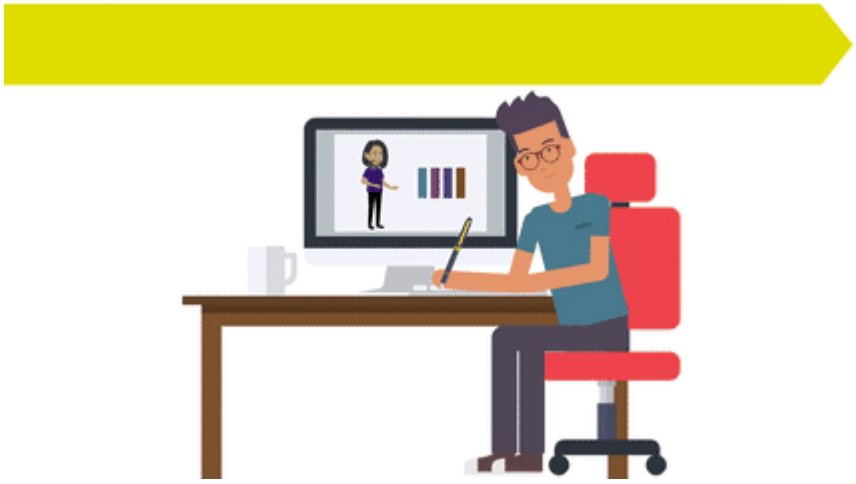
Lesson 9 was solely concerned with Hierarchical clustering methods. The first topic gave a description of what hierarchical clustering is and when such an approach would be used, i.e., within an un-supervised learning study. We also discussed the differences between hierarchical and non-hierarchical clustering, plus run through a simple coded example. Topic 2 ran through the Agglomerative clustering method, where it takes each object in the data set to be a single cluster. All the clusters are then merged together in a step-by-step manner, until only one cluster remains, achieved through proximity values of the individual data points.

In Topic 3 we looked at the divisive method, also known as the top-down approach, opposite to the Agglomerative method, and in the final topic we culminated everything that you had learned in a full example.

Continue

Topic 1 - Activity 1

Make a Note



Write a brief overview of what you have learned personally within the module, in its entirety. You can be as brief, or in-depth as you like. When you have completed your overview, upload the document to the group's discussion board.

You can use the below Notepad to help you with this activity, but feel free to use whichever note-taking tool you prefer.

If you decide to use the below Notepad, please remember anything you type in the box below will not be kept when you leave the lesson, so you will need to download your notes for revision and discussion purposes.

My Notes

Input Your Notes Here

Continue

TOPIC 1 SUMMARY

By consolidating the weekly lessons in short summaries, hopefully you can see the long and tough journey you have completed, especially if this is the first time that you have engaged in and Machine Learning. If you have found this module difficult, then this is perfectly normal: new experiences and activities usually are. However, you have battled through and found yourselves knowing much more about the coded environment that makes most models possible within a data science environment, than you did at the beginning of the module. The algorithms do take a little getting used to, but with practice, practice and more practice, you will no doubt continue the machine learning enlightenment you are now on.

Continue to Topic 2

Topic 2: Assignment support



Image from Articulate Rise 360 Content Library.

As you are hopefully aware, there are many ways in which both your tutor and Arden University can support assignment writing and student life in general. Whether that be guidance on academic conventions for assignment writing, or support if you are finding student life a little difficult. Your tutor should be the first person you approach if you feel you need additional support in any aspect of university life. The tutor will discuss any concerns with you and offer guidance wherever possible. Of course, any discussions had with your tutor, or Arden university staff will be totally confidential. If third parties have been identified for additional support by

your tutor within Arden University, then permissions by the student will be requested before any third parties are involved from the university.

There are many ways in which you can contact your tutor, you can post a message in your module's discussion forum or contact your tutor by email, to name but a few. In addition to this, your tutors' information can be found on the right-hand side of your module page. Please find below a list of support services which can be found at Arden university, with brief descriptions, to help you have a great students experience.

For general support and guidance:

Email: studentsupport@arden.ac.uk

Phone: +44 2476 515700

If you are having problems accessing your content materials or anything technical in nature, please contact the friendly technical support team who will be happy to assist.

Email: help@arden.ac.uk

The Academic Skills team offers a plethora of support in all aspects of academic conventions, including referencing, accessing materials, reading, critical thinking and literature searching. You can book a one-to-one session or attend any of the support sessions the team run throughout the academic calendar.

Website: Academic Skills Team

Email: study-skills@arden.ac.uk

The Arden Library offers a wide and extensive range of literature to help you with your studies. Whether that be books, subject guides, journal articles for that deeper look at your module's topic, or blogs. You will find what you are looking for in Arden University's Library Portal.

Website: Library Portal

Email: libraries@arden.ac.uk

Everyone finds life and studies overwhelming at times. Arden University recognises this and provides excellent additional support in this area.

Website: Wellbeing Services Student Portal

Here at Arden, we want to ensure that everyone enjoys themselves and has a great learning experience. If you require any help or support while studying at Arden, there is a wealth of experience within each department to help, support and guide you through, whatever difficulties you may be experiencing.

Continue

Topic 2 - Activity 1

Arden support



Please ensure that you familiarise yourselves with the support available to you at Arden and send an introductory email to each department. This is a good way of not only recognising where the support is available to you, but also each team will know you are studying at Arden.

Continue

TOPIC 2 SUMMARY

We have now revisited many of the support centres that are available to you at Arden University. Please do not feel that you are isolated with your time at Arden because you are not. If you do need any support throughout your studies, please have an initial discussion with your tutor.

Continue

Did you know?

Arden University was highly commended at the Digital Education Awards 2023 (Arden University, 2023b).



Image from Articulate Rise 360 Content Library.





Image from Articulate Rise 360 Content Library.

Arden is also the name of a village in Delaware (VisitDelaware.com, 2025)

Continue to Lesson summary

Lesson summary

LESSON SUMMARY

Congratulations! You have not only completed Lesson 10, but you have also completed the entire Machine Learning module.

If this is the first time that you have engaged in any type of machine learning, then I am sure you will agree that you found some aspects of the module a little more difficult than others. But nevertheless, you have completed it. Hopefully, you have expanded your knowledge on how to apply many Python libraries and you will no doubt, initially, refer to this module and its content many times as you progress forward in your degree and professional career.

Lesson 10 has been concerned with reflection and additional support to help guide your through not only this module, but your degree in its entirety. We summarised each lesson, so that there is a brief overview of what each lesson contains, for your reference. Importantly, the many different areas in which support is available to you throughout your studies at Arden have been detailed and links given.

Continue to 'Reading list'

Reading List

All reading and resources in this module are provided by the Arden University Library service. The full reading list for this module can be accessed at the link below. More books, journals and databases can be found on the Library Portal. If you need these resources in a different format, our accessibility tools can help.

Full reading list

You can find all the key, essential and wider reading and resources for this module on the dedicated reading list page on the Learning Success Hub.

VIEW READING LIST

References

Cited sources that do not appear in your reading list:

Arden University, (2023a). *Arden nominated for seven EducationInvestor Awards 2023 | Arden University* [webpage, online]. Arden University.

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